THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE CHAPTER TEST SET B

Chapter 7: **Measurement of Time and Motion** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A bus covers 150 km in 3 hours. Its average speed is: 1

(a) 25 km/h **(b)** 60 km/h **(c)** 450 km/h **(d)** 50 km/h

2. A bus covers 75 km in 1.5 hours. Its average speed is: 1

(a) 112.5 km/h **(b)** 25 km/h **(c)** 50 km/h **(d)** 60 km/h

3. A car moves at a constant 45 km/h for 2 hours. The distance covered is: 1

(a) 45 km **(b)** 22.5 km **(c)** 135 km **(d)** 90 km

4. A cyclist travels 100 km at 25 km/h. The time taken is: 1

(a) 4 hours **(b)** 5 hours **(c)** 8 hours **(d)** 2500 hours

5. 36 km/h is equal to: 1

(a) 0.6 m/s **(b)** 10 m/s **(c)** 129.6 m/s **(d)** 36 m/s

6. 10 m/s is equal to: 1

(a) 100 km/h **(b)** 600 km/h **(c)** 2.8 km/h **(d)** 36 km/h

7. The SI unit of time is the: 1

(a) minute **(b)** hour **(c)** second **(d)** day

8. A pendulum clock uses the: 1

(a) sand flow **(b)** water flow
(c) regular oscillation of a pendulum **(d)** Sun's shadow

9. The speed of an object is: 1

(a) force per unit area **(b)** mass per unit volume
(c) time per unit distance **(d)** distance covered per unit time

10. An object covering unequal distances in equal intervals has: 1

(a) uniform motion **(b)** non-uniform motion
(c) zero speed **(d)** no motion

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 15–25 words each.

11. What is a sundial? Why cannot it work at night? 2

12. A car covers 150 km in 3 hours. Find its speed. 2

13. Convert 72 km/h to m/s. 2

14. What is a clepsydra? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Distinguish between speed and distance. 3

16. A pendulum has a time period of 1.5 s. How many oscillations in 1 minute? 3

17. State two limitations of a sundial and how a clock overcomes them. 3

OR

Why are standard units needed for measurement of time? 3

18. Convert 90 km/h into m/s and 8 m/s into km/h. 3

OR

An athlete runs 400 m in 50 s. Find the speed in m/s and km/h. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–120 words each.

19. (i) Distinguish between uniform and non-uniform motion with examples. 5
(ii) Draw distance–time graph shapes (describe).
(iii) Why is average speed needed?

OR

(i) Convert 108 km/h and 25 m/s. 5
(ii) Which is faster?
(iii) How far in 10 s at the faster speed?

20. **Case study:** A runner covers 100 m in 12.5 s. 5
(i) Find speed in m/s.
(ii) Convert to km/h.
(iii) Time for 400 m at same speed.
(iv) Which instrument measures it?

OR

(i) What are the units of time used earlier in India? 5
(ii) How were they measured?
(iii) Why standardise?

END OF QUESTION PAPER — Total: 40 Marks

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